

HR Analytics Dashboard – Final Project

Excel Data Analytics Project

1. Project Overview

This project analyzes employee data using Microsoft Excel to understand workforce composition, employee attrition, and differences across departments, cities, job roles, education levels, gender, and age groups. The final output is an interactive HR Analytics Dashboard using KPI cards, PivotTables, PivotCharts, and slicers.

2. Key KPIs

Total Employees: 1,000 **Attrition Count:** 508 **Active Employees:** 492 **Attrition Rate:** 50.8% **Average Age:** 41

3. Dashboard Output

The dashboard below is the final visual output of the project. It combines workforce KPIs, gender and education analysis, job-role distribution, department attrition, age-wise attrition, city-wise attrition, and interactive Department and Education slicers.

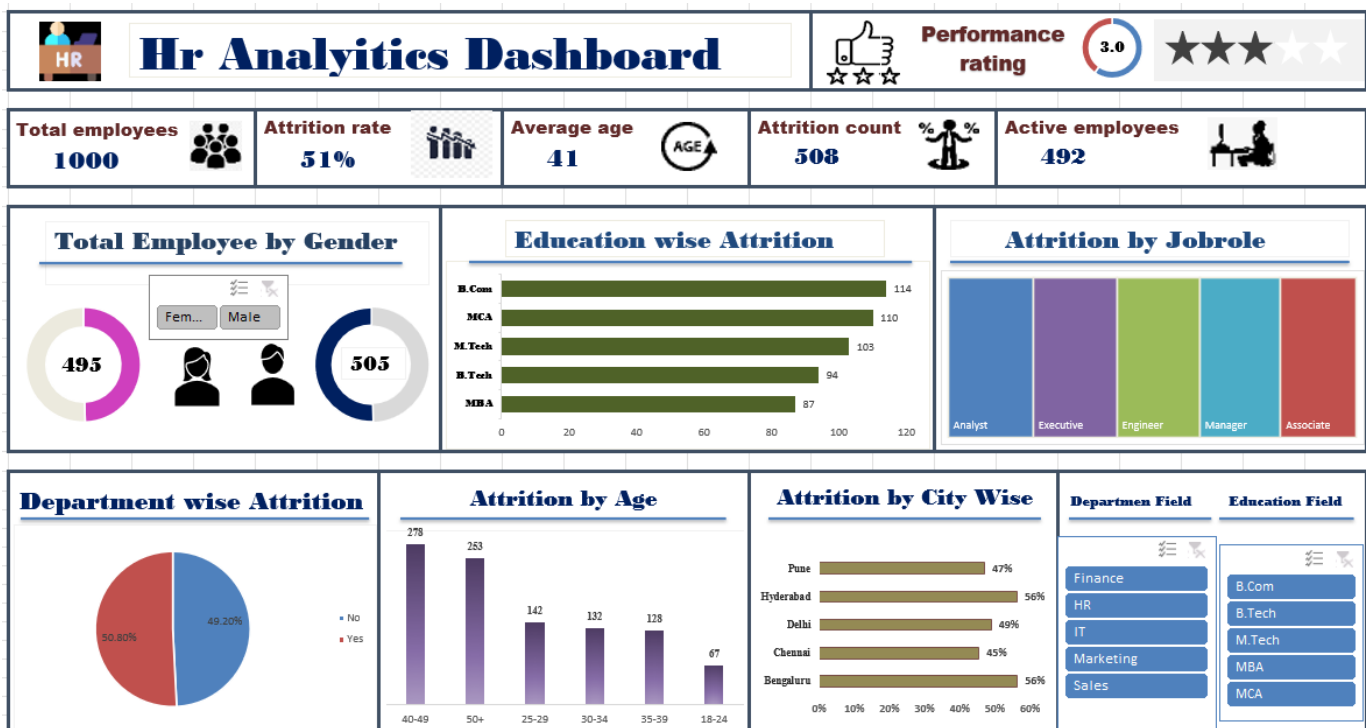


Figure 1. Final HR Analytics Dashboard output.

4. Data Cleaning

The data preparation process included checking missing values, reviewing duplicate records, checking invalid or negative salary values, standardizing inconsistent text values such as city, department, education, gender, and job-role labels, and converting the cleaned dataset into an Excel Table.

5. Analysis Performed

The dashboard supports analysis of employee count by gender, attrition by education, job-role distribution, department-wise attrition, age-group attrition, and city-wise attrition. Slicers provide interactive filtering for Department and Education.

6. Important Metric Definition

Department Attrition Rate = Employees with Attrition = Yes in the department ÷ Total employees in that department. This is a department-level rate and should not be confused with the department's percentage of grand total.

7. Business Conclusion

The analysis shows relatively high overall attrition in the dataset and variation across workforce segments. The dashboard helps HR identify departments, cities, age groups, education groups, and job roles that deserve closer investigation. The analysis is descriptive; it does not by itself prove that any specific factor causes attrition. Further analysis of salary, overtime, experience, job role, performance, and other employee attributes would be required before taking targeted retention actions.

8. Future Improvements

Future versions can use Power Query for automated data cleaning, Power Pivot/Data Model with DAX for scalable measures, and Power BI for more advanced interactive reporting. Additional analysis could also investigate relationships between attrition and salary, overtime, tenure, performance, and job role.

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